

Title

Estimation Limits of Atmospheric Monitoring with Integrated Sensing and Communication in Sub-THz Non-Terrestrial Networks

Abstract

This topic presents an estimation framework for opportunistic atmospheric sensing using sub-Terahertz (60–400 GHz) Non-Terrestrial Network downlinks. The proposed system aims to simultaneously estimate the presence of atmospheric pollutants and quantify it via assessing densities of particulate matter (PM) and gaseous volatile organic compounds (VOCs).

The solution uses two distinct physical phenomena sensed from 6G Channel State Information (CSI) amplitudes: discrete rotational molecular absorption notches (simulated effect of VOCs with databases such as HITRAN or similar) and the smooth Rayleigh scattering baseline (for PM of varying sizes). To establish the viability of this architecture, the research will define the fundamental limits of pollutant detection under satellite link noise constraints. This work will provide a foundation for satellite-based environmental monitoring without degrading communication capacity. This can convert communication networks into global environment monitoring networks providing data to climatologists and emergency services (wildfire monitoring and response).

Research Tasks

Task 1: Stratified Atmospheric Channel Modeling

- **Task 1.1:** Develop a multi-layer tropospheric profile where pressure, temperature, and molecular density vary with altitude based on standard atmospheric reference models.
- **Task 1.2:** Use the HITRAN spectroscopic database to extract line positions, intensities, and broadening coefficients for target rotational gas transitions within the targetted frequency window.
- **Task 1.3:** Implement an atmospheric line-shape profile (Voigt/Lorentz) to integrate molecular absorption along a vertical slant path determined by the satellite elevation angle.
- **Task 1.4:** Model the frequency-dependent Rayleigh aerosol scattering baseline for particulate matter across the same stratified slant path.

Task 2: NTN Link Parameterization and CSI Synthesis

- **Task 2.1:** Define the link budget and physical parameters for a 6G sub-THz satellite downlink (e.g., carrier frequency, wideband OFDM subcarrier spacing, transmit power, and high-gain directional receiver antenna profile).
- **Task 2.2:** Simulate the received wideband CSI, modeling a Line-of-Sight channel under the integrated slant-path atmospheric attenuation and Additive White Gaussian Noise (AWGN).

Task 3: Feature Inversion and Parameter Extraction

- **Task 3.1:** Design an estimation pipeline to isolate the spectral notches (gas lines) from the broad frequency-dependent attenuation caused by PM.
- **Task 3.2:** Formulate and evaluate algorithmic approaches (such as optimization-based curve fitting, statistical ratiometric inversion, or machine learning estimators) to extract the gas concentration and PM density values from the CSI.

Task 4: Performance Bounding and Sensitivity Evaluation

- **Task 4.1:** Derive the analytical lower bound of estimation variance given the link's SNR.
- **Task 4.2:** Conduct sensitivity analyses to evaluate estimation error and lower bounds as a function of changing satellite elevation angles and varying atmospheric noise levels.
- **Task 4.3:** Quantify the exact parts-per-million (ppm) and mass density detection floors of the developed estimator to verify compliance with environmental sensing standards..

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